



The Indian international School, DSO

Chemistry Department

PRACTICAL

Systematic Quantitative Analysis (Volumetric Analysis)

Estimate the strength and molarity of the given KMnO_4 by titrating it against M/20 Mohr's salt solution.

AIM: Estimate the molarity and strength of the given KMnO_4 by titrating it against M/20 or (0.05M) Mohr's salt.

PRINCIPLE:

Mohr's salt being a primary standard substance, it is used for standardization of KMnO_4 solution. In acidic medium Mohr salt is oxidized by KMnO_4 as -

Ionic equation-

Oxidation half reaction - $[\text{Fe}^{2+} \rightarrow \text{Fe}^{3+} - \text{e}^-] \times 5$

Reduction half reaction - $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$

Overall ionic equation - $\text{MnO}_4^- + 8\text{H}^+ + 5\text{Fe}^{2+} \rightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$

Burette solution - KMnO_4

Pipette solution- Mohr's salt.(20 ml)

Indicator - Self indicator, KMnO_4

End point : Colour change from colourless to light pink.

PROCEDURE:

(I) PREPARATION OF M/20 FERROUS AMMONIUM SULPHATE (FAS) SOLUTION

Add 10 ml conc. H_2SO_4 in a 100mL standard flask and mix 1.96 gm FAS in it. After that add distilled water to make up the solution to 100mL. Shake well to get a uniform solution.

II) TITRATION:

- (1) Rinse and fill the burette with KMnO_4 solution and remove air bubble from the nozzle.
- (2) Suck out 20 ml of FAS by using pipette and transfer it into conical flask.
- (3) Add half test tube dil. H_2SO_4 into conical flask and place it below burette.
- (4) Add KMnO_4 into conical flask drop by drop and stir well.

(5) At first solution turns pink which on shaking disappears. Repeat the process until the solution becomes stable pink. Note down the reading of the burette.

(6) Repeat the process to get concordant values

THE FOLLOWING WRITE (LEFT HAND SIDE)

Observations and calculations: –

Calculation - Weight of Mohr's salt.:

$$W_2 = \frac{M \times M_2 \times V}{1000}$$

$$W_2 = \frac{0.05 \times 392 \times 100}{1000}$$

$$= 1.96 \text{ g}$$

M- Molarity of Mohr's salt= 0.05 M

M2- Molar mass of Mohr's salt = 392 g

V- Volume = 100ml

S. No.	Volume of Mohr's salt(ml)	Burette reading		Concordant value	Volume of KMnO ₄ (ml)
		Initial	Final		
1.	20	0			-----ml
2.	20	0			
3.	20	0			

a) Molarity of KMnO₄ :-

$$M_1V_1 = n_1 (\text{KMnO}_4)$$

$$M_2V_2 = n_2 (\text{Mohr's salt})$$

$$M_1V_1n_2 = M_2V_2n_1$$

M₁- Molarity of KMnO₄ ??

V₁- Volume of KMnO₄

n₁- Number of moles of KMnO₄ =1

M₂- Molarity of Mohr's salt= 0.05 M

V₂- Volume of Mohr's salt = 20 ml

n₂- Number of moles of Mohr's salt=5

Molarity of KMnO_4 , M_1

$$M_1 = \frac{M_2 V_2 n_1}{V_1 n_2} = \frac{0.05 \times 20 \times 1}{V_1 \times 5} = \text{-----} M$$

b) **Strength of KMnO_4** = Molarity of KMnO_4 x molecular weight of KMnO_4
= ----- x 158 = -----g/l

RESULT –

a) Molarity of given KMnO_4 = ----- M

b) Strength of given KMnO_4 = ----- g/l

Reference:

[Comprehensive Practical Chemistry Xii Cbse | PDF A Handbook of Practical Chemistry Class XII - Dr M L Sharma, Dr P N Chaudhary, Dr P S Saud | Thuprai](#)